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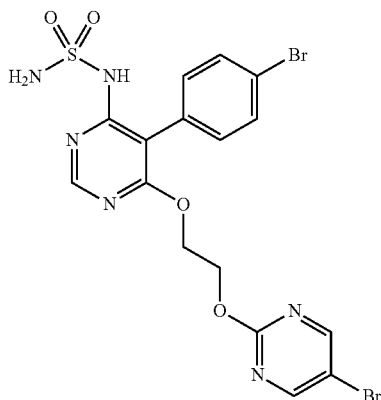
**CRYSTALLINE FORMS OF A
4-PYRIMIDINESULFAMIDE DERIVATIVE
APROCITENTAN**

**CROSS REFERENCE TO RELATED
APPLICATIONS**

This application is a continuation of U.S. patent application Ser. No. 16/489,194 filed Aug. 27, 2019, which is a National Stage Entry under 35 U.S.C. § 371 claiming benefit of PCT Application No. PCT/EP2018/054627, filed on Feb. 26, 2018, which claims the benefit of PCT Application Nos. PCT/EP2017/054489, filed on Feb. 27, 2017, and PCT/EP2017/061487, filed on May 12, 2017, the contents of each of which are incorporated herein by reference.

The present invention concerns novel crystalline forms of {5-(4-bromo-phenyl)-6-[2-(5-bromo-pyrimidin-2-yloxy)-ethoxy]-pyrimidin-4-yl}-sulfamide (hereinafter also referred to as "COMPOUND"), processes for the preparation thereof, pharmaceutical compositions comprising said crystalline forms, pharmaceutical compositions prepared from such crystalline forms, and their use as endothelin receptor inhibitors and endothelin receptor antagonists. It also relates to uses of the COMPOUND for treating particular diseases or disorders, either alone or in combination with other active ingredients or therapeutic agents.

Aprocitentan, {5-(4-bromo-phenyl)-6-[2-(5-bromo-pyrimidin-2-yloxy)-ethoxy]-pyrimidin-4-yl}-sulfamide has the formula I



Aprocitentan, also known under the name ACT-132577, is an endothelin receptor inhibitor and useful as endothelin receptor antagonist. The compound of formula I is a member of a structural family that was previously generically disclosed in WO 02/053557. In particular, the compound of formula I, while showing endothelin receptor antagonist activity, exhibits in vivo a much longer half-life and a much shorter clearance in comparison to corresponding alkylated derivatives. This makes the compound of formula I particularly suitable for long-acting pharmaceutical compositions, as disclosed in WO 2009/024906. Certain manufacturing processes relating to aprocitentan are disclosed in WO2015/121397.

Because of its ability to inhibit the endothelin binding, COMPOUND can be used for treatment of endothelin related diseases which are associated with an increase in vasoconstriction, proliferation or inflammation due to endothelin. Examples of such endothelin related diseases are

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hypertension, pulmonary hypertension, coronary diseases, cardiac insufficiency, renal and myocardial ischemia, renal failure, cerebral ischemia, dementia, migraine, subarachnoidal hemorrhage, Raynaud's syndrome, digital ulcers and portal hypertension. They can also be used in the treatment or prevention of chronic kidney disease (CKD), diabetes, diabetic nephropathy, diabetic retinopathy, diabetic vasculopathy, chronic heart failure and diastolic dysfunction. they can further be used in the treatment or prevention of atherosclerosis, restenosis after balloon or stent angioplasty, inflammation, stomach and duodenal ulcer, cancer, melanoma, prostate cancer, prostatic hypertrophy, erectile dysfunction, hearing loss, amaurosis, chronic bronchitis, asthma, pulmonary fibrosis, gram negative septicemia, shock, sickle cell anemia, glomerulonephritis, renal colic, glaucoma, connective tissue diseases, therapy and prophylaxis of diabetic complications, complications of vascular or cardiac surgery or after organ transplantation, complications of cyclosporin treatment, pain, hyperlipidemia as well as other diseases, presently known to be related to endothelin.

According to the 2014 American Society of Hypertension and International Society of Hypertension joint statement [Weber et al., "Clinical Practice Guidelines for the Management of Hypertension in the Community. A Statement by the American Society of Hypertension and the International Society of Hypertension." J Clin Hypertens (2014), 16(1), 14-26], the 2013 European Society of Hypertension and European Society of Cardiology joint guideline [Mancia et al, J. Hypertens. (2013), 31, 1281-1357], as well as several national guidelines [Denolle et al., J Hum Hypertens. (2016), 30(11), 657-663; McCormack et al., Br J Cardiol (2013), 20 (suppl 1), S1-S16], resistant hypertension (rHT) (or difficult to treat hypertension) is defined as uncontrolled blood pressure (BP) (i.e., failure to lower BP to a pre-defined threshold) despite concurrent administration of three anti-hypertensive therapies of different pharmacological classes at maximal or optimal doses, including a diuretic. Thus, resistant hypertension patients include patients whose blood pressure is controlled with use of more than three medications. That is, patients whose blood pressure is controlled but require four or more medications to do so should be considered resistant to treatment (see e.g. Mancia et al, J. Hypertens. (2013)).

Clinical studies have shown that endothelin receptor antagonists (ERAs) may have significant treatment effect in patients suffering from hypertension and/or renal disease. However, therapeutic benefit needs to be weighted against potential side effects, such as the potential risk of teratogenic activity. In addition, both, selective ETA-antagonists and dual antagonists of both the ET_A and ET_B receptor, may cause fluid retention, a common side effect associated with many previously studied ERAs and sometimes (e.g. if not manageable with diuretics) leading to exaggerated major adverse cardiac events such as heart failure or death. Whereas the risk-benefit balance is in most cases in favor of treatment with an ERA for indications such as pulmonary hypertension (as reflected in the past by successive market approvals e.g. for the ERAs the dual antagonists bosentan and macitentan, the ET_A-selective antagonist ambrisentan), ERAs have no role in the management of primary hypertension (Laffin et al. Seminars in Nephrology 2015, 35, 168-175), and side effects such as fluid retention may remain an issue when a potential treatment of rHT, chronic kidney disease or other hypertension related diseases with an ERA is considered.

The ET_A-selective endothelin receptor antagonist darusentan has been in development for the treatment of rHT

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